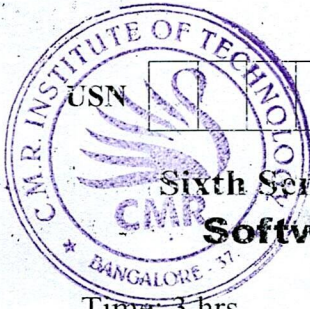




Sixth Semester B.E./B.Tech. Degree Examination, June/July 2025

Software Engineering and Project Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is a software? What are the characteristics of software that is different from hardware? (08 Marks)
- b. Demonstrate the role of each activity in a generic process framework for software engineering. (08 Marks)
- c. Software engineering is a layered technology. Justify with necessary diagram and explanation. (04 Marks)

OR

- 2 a. Explain waterfall model along with its variation with neat diagram. (08 Marks)
- b. Interpret the seven general principles that focus on software engineering. (08 Marks)
- c. Justify that prototyping serves as the better mechanism for identifying software requirements. (04 Marks)

Module-2

- 3 a. Describe the distinct tasks of requirement engineering. (08 Marks)
- b. Sketch an activity diagram for access camera surveillance via the internet-display camera views function. (08 Marks)
- c. Show how the requirements can be validated. (04 Marks)

OR

- 4 a. Discuss the concepts of data modeling with examples. (08 Marks)
- b. Demonstrate the use of CRC modelling in system requirements with an example. (08 Marks)
- c. Distinguish among association, multiplicity and dependencies. (04 Marks)

Module-3

- 5 a. What is Agility? Explain agility principles to be followed to achieve agility. (10 Marks)
- b. Demonstrate extreme programming with diagram and give its values. (10 Marks)

OR

- 6 a. Describe adaptive software development and scrum agile process models. (10 Marks)
- b. Interpret the principles of framework activity "communication". (10 Marks)

Module-4

- 7 a. What are activities covered by software project management? (06 Marks)
- b. Analyze the software development life cycle with reference to ISO 12207. (10 Marks)
- c. Discuss W5HH principle. (04 Marks)

OR

- 8 a. Discuss the process of setting objectives for project. (06 Marks)
- b. Examine the activities involved in software project management and functioning of the project control cycle. (10 Marks)
- c. Differentiate between software development and project management life cycles. (04 Marks)

Module-5

- 9 a. Demonstrate the place of software quality in stepwise framework with neat diagram. (10 Marks)
- b. Distinguish between Dromey's quality model and Boehm's quality model. (10 Marks)

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- 10 a. Give an insight on ISO 9126 with reference to software quality. (10 Marks)
- b. Analyze the role of Capability Maturity Model (CMM) to improve quality of software development. (10 Marks)

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